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Reinforcement Learning-Based Vehicle Autonomous Emergency Braking System

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강화학습 기반 차량 자율 긴급 제동 시스템

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ABSTRACT

This study proposed a reinforcement learning-based autonomous emergency braking (AEB) system designed for complex and unpredictable intersection scenarios. To address the limitations of rule-based approaches in dynamic traffic environments, a deep deterministic policy gradient algorithm was employed to learn continuous braking control policies. A simulation environment was established by integrating IPG CarMaker with MATLAB/Simulink, and sensor data from LiDAR and radar were used to define the critical state variables. The reward function was designed to balance collision avoidance with driving efficiency by penalizing unnecessary deceleration and rewarding safe distance maintenance. Three intersection scenarios9-left-turn, straight-crossing, and right-turn violations-were implemented to evaluate the system. The simulation results demonstrated that the proposed method achieved stable policy convergence across all scenarios and successfully avoided collisions at various initial speeds (60–90 km/h). These results confirm that the reinforcement learning-based AEB system provides consistent and reliable braking performance, suggesting its potential applicability to broader autonomous driving environments.

Keywords : Autonomous Driving(자율주행), Virtual Environment(가상 환경), Emergency Braking(긴급 제동), Reinforcement Learning(강화학습)

1. Introduction

Autonomous driving technology has been rapidly evolving in recent years with the advancement of artificial intelligence, and it has emerged as one of the core technologies for future transportation systems. This technology provides various social benefits, including traffic congestion reduction, improved energy efficiency, enhanced mobility for the elderly and transportation vulnerable groups, and a decrease in traffic accidents. In particular, since a significant portion of traffic accidents are caused by human negligence or judgment errors, there is a growing demand for autonomous driving systems capable of perceiving, reasoning, and controlling vehicle behavior. An autonomous driving system replaces human cognitive, decision making, and control functions with artificial intelligence (AI), and to achieve this, it requires the integration of various sensors and AI-based algorithms. However, roads shared by both autonomous and human-driven vehicles are highly

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complex and unpredictable, where irregular behaviors of surrounding vehicles and sensor uncertainties coexist. In such conditions, ensuring both safety and efficiency in driving decision-making remains challenging. Conventional rule-based autonomous driving systems make decisions and control actions based on predefined conditions and logic structures, which limits their flexibility in unpredictable situations and increases the potential for malfunction. Particularly at intersections where interactions between multiple agents are frequent and complex rule-based approaches often fail to effectively balance between conservative and delayed braking behaviors[1]. To overcome these limitations, recent studies have focused on the application of artificial intelligence technologies such as deep learning[2]. Reinforcement learning (RL), in particular, is well-suited for decision-making in uncertain and dynamic traffic environments, as it enables an agent to learn optimal policies that maximize long-term rewards through interaction with the environment. Ye et al. [3] applied the Deep Q-Network (DQN) algorithm to train autonomous vehicles to avoid collisions with pedestrians, while Qi et al. [4] achieved energy-efficient driving control using the Deep Deterministic Policy Gradient (DDPG) algorithm. Other studies have introduced RL-based methods for specific driving scenarios such as vehicle following, lane keeping, and highway driving[5]~[8]. Nevertheless, most existing studies have focused on limited or single-objective scenarios, and systematic validation from a continuous control perspective including emergency braking, collision avoidance, and resumption of driving in intersection environments remains insufficient.

Therefore, this study aims to develop and verify a reinforcement learning based autonomous emergency braking (AEB) system for unexpected situations that may occur at intersections. The DDPG algorithm, which is suitable for continuous action spaces, was adopted, and an integrated simulation environment was established using IPG CarMaker and MATLAB/Simulink. The state variables were defined using key information such as the relative distance and relative velocity obtained from LiDAR and radar sensors mounted on the vehicle's front. The reward function was designed to balance collision avoidance and minimal deceleration, while termination conditions were structured to allow smooth braking release and resumption of driving once the danger was resolved. Finally, the generalization performance of the trained policy was evaluated through various speed ranges and three intersection scenarios.

2. Experimental Configuration

2.1 System Configuration

To implement the autonomous emergency braking system, a virtual driving environment was constructed using the vehicle dynamics simulation software IPG CarMaker. MATLAB/Simulink was integrated with CarMaker to design and train both the vehicle control logic and the reinforcement learning agent. The overall system architecture is illustrated in Figure 1.

The vehicle model used in the experiment was a BMW 5 Series. As shown in Figure 2, in order to perceive surrounding obstacles and driving conditions, LiDAR and radar sensors were mounted on the front of the vehicle. The LiDAR provides distance and shape information of the objects ahead, while the radar measures their relative velocity and distance. Sensor data are transmitted to the reinforcement learning agent in real time, and the agent computes the corresponding state and reward values. Through the training process, the agent learns the optimal braking policy and generates appropriate braking commands according to the driving situation. The sensor parameters were configured using the default settings provided in CarMaker.

Fig. 1 System configurations

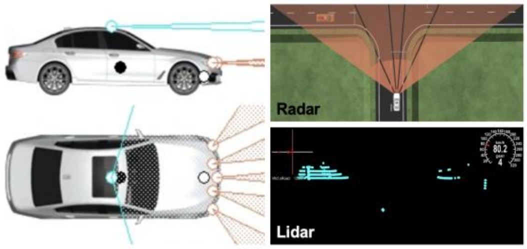
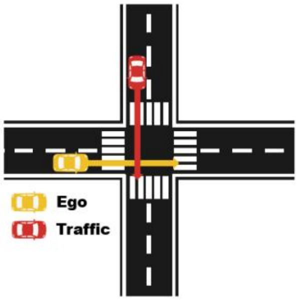
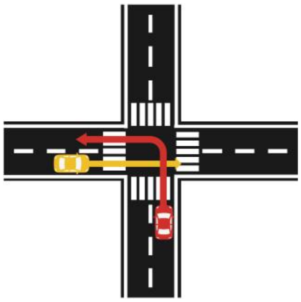


Fig. 2 Sensor overview



(a) SCP/L

(b) LTAP/LD



(c) RTAP

Fig. 3 Road scenarios

2.2 Scenario Definition

Among autonomous driving scenarios, intersections are considered one of the most complex and challenging environments due to their unpredictable and dynamic traffic conditions. Previous studies have pointed out that autonomous emergency braking systems often exhibit limited detection accuracy for unexpected events occurring from lateral directions. To address this limitation, this study designed multiple emergency braking scenarios that include various unexpected events at intersections. A total of three intersection scenarios were developed, as illustrated in Figure 3. The ego vehicle was assumed to proceed straight through the intersection according to the traffic signal, while the opposing vehicles were set to violate traffic rules or fail to yield, thereby creating potential collision risks. Specifically, Figure 3(a) represents a scenario where a vehicle from the left proceeds straight through a red light, Figure 3(b) depicts a vehicle turning left from the right side while ignoring the signal, and Figure 3(c) describes a right-turning vehicle entering the intersection without checking for oncoming traffic. All three scenarios were implemented in IPG CarMaker to closely replicate real intersection conditions and were used to train and evaluate the performance of the proposed reinforcement learning–based autonomous emergency braking model.

2.3 DDPG Algorithm

Reinforcement learning is a machine learning technique in which an agent interacts with an environment and learns a policy that maximizes cumulative rewards through trial and error. In the conventional Q-learning method, the value of each state-action pair is stored in a Q-table. However, as the state space becomes high-dimensional, the memory requirements and computational complexity increase exponentially. To address this issue, the DQN was proposed, which approximates the Q-function using a deep neural network and has demonstrated effective performance in discrete action spaces. However, DQN is difficult to apply directly to continuous control problems due to the so-called “curse of dimensionality.” To overcome this limitation, the Deterministic Policy Gradient (DPG) method was introduced. Unlike stochastic policy approaches, DPG deterministically outputs specific actions, making it suitable for continuous control tasks. The Deep Deterministic Policy Gradient (DDPG) algorithm combines the advantages of DQN and DPG. It adopts an actor-critic architecture and utilizes a replay buffer and target networks to enhance learning stability. The actor network outputs continuous control inputs from the given state, while the critic network estimates the value of the corresponding action as a Q-value. In this study, the DDPG algorithm was selected because autonomous braking requires continuous-valued control inputs, and deterministic actor-critic methods are well suited for learning smooth and fine-grained pedal modulation. Therefore, DDPG provides an appropriate framework for modeling realistic braking behavior in continuous action spaces. The proposed reinforcement learning model is presented in Figure 4 and the environment is composed of roads and vehicles, and the vehicle control device is modeled as an agent.

The architecture of the proposed DDPG model is presented in Figure 5. In this study, the DDPG framework was implemented using the reinforcement learning agent block in MATLAB/Simulink and integrated with IPG CarMaker to receive real-time vehicle dynamics and sensor data. The observation variables were defined to reflect both the dynamic state of the ego vehicle and its interaction with surrounding vehicles. These variables include the brake pedal input, ego vehicle velocity and acceleration, LiDAR-based distance to the preceding object, radar-based relative distance and relative velocity, simulation time, and collision state, as summarized in Table 1. The action space was defined as the brake pedal input, represented as a continuous control value ranging from 0 to 1, where 0 corresponds to no braking and 1 represents maximum braking. The reward function was designed to simultaneously consider driving safety and efficiency. A large negative reward was assigned in the event of a collision, and a penalty was applied when the vehicle approached a dangerous distance. Conversely, positive rewards were given for maintaining safe distances, while unnecessary early braking and excessive deceleration were penalized. To ensure smooth braking behavior, the rate of change in acceleration was also included as part of the reward structure. The overall reward at each time step R_t was computed as follows:

$$\begin{aligned} R_t &= r_{LiDAR} + r_{time} + r_{collision} + r_{distance} \\ &\quad + r_{early} + r_{emergency} + r_{radar} \end{aligned} \tag{1}$$

and each term is defined as follows.

$$r_{LiDAR} = 0.8 \cdot \mathbb{I}(d_{LiDAR} > 300) \tag{2}$$

$$r_{time} = t \tag{3}$$

$$r_{collision} = \begin{cases} -5000, & if\ collision\ occurs \\ 0, & otherwise \end{cases} \tag{4}$$

$$r_{distance} = 1.5 \cdot \left(d_{car} - d_{ref}\right) \tag{5}$$

$$r_{early} = -80 \cdot \mathbb{I}(brake\ too\ early) \tag{6}$$

$$r_{emergency} = -30 \cdot \mathbb{I}(stop\ triggered) \tag{7}$$

$$r_{radar} = 1.5 \cdot v_{rel} \tag{8}$$

The designed reward structure aims to encourage safe, efficient, and comfortable braking behavior under uncertain traffic conditions. The LiDAR-based distance term r_{LiDAR} rewards the agent for maintaining sufficient clearance from the preceding vehicle, ensuring early hazard perception. The time-based term r_{time} encourages the agent to complete the driving task within a reasonable time, preventing overly conservative braking. The collision term $r_{\mathrm{collision}}$ provides a large negative reward upon any collision, explicitly guiding the policy to prioritize safety. The distance term r_{distance} reflects the vehicle's relative position, preventing tailgating and promoting safe following behavior. The early braking term r_{early} penalizes unnecessary braking when the target vehicle is still distant, improving ride comfort and fuel efficiency. The emergency stop term $r_{\mathrm{emergency}}$ penalizes sudden or full-brake actions that may lead to discomfort or instability. Lastly, the radar-based velocity term r_{radar} ensures that braking intensity is properly adapted to the relative speed of the obstacle. Together, these reward components enable the agent to learn a policy that reacts rapidly to sudden obstacles while maintaining smooth and stable braking under various intersection conditions. The constant coefficients used in the reward function were determined through an iterative adjustment process during training. After defining each reward component, the agent was trained while continuously monitoring the intermediate reward values, braking responses, and convergence behavior. When certain reward terms produced disproportionately large or small values, the corresponding coefficients were modified to achieve a balanced influence across all components. The collision penalty was intentionally set to the largest negative value because the agent frequently failed to prioritize safety when the penalty magnitude was small. By increasing the collision penalty, the agent reliably learned to avoid trajectories that led to impact. For the distance-related, early-braking, emergency stop, and relative-velocity terms, the coefficients were gradually tuned by observing how the agent reacted in real time. If the agent braked unnecessarily early, the early-braking penalty was strengthened; if braking was too late or overly aggressive, the distance and emergency stop coefficients were adjusted. Because each reward component has a different numerical scale, the coefficients were refined so that no single term dominated the reward excessively unless intentionally designed to do so. Through repeated simulation trials, the final coefficient values were set at levels that produced stable learning behavior, smooth braking responses, and reliable collision avoidance across scenarios. The termination conditions were defined such that the simulation ended immediately upon collision or when the predefined maximum simulation time was exceeded. Training was conducted independently for each of the three intersection scenarios, and the learned policies were then evaluated within their respective environments to verify braking performance. The hyperparameters used in training are listed in Table 2.

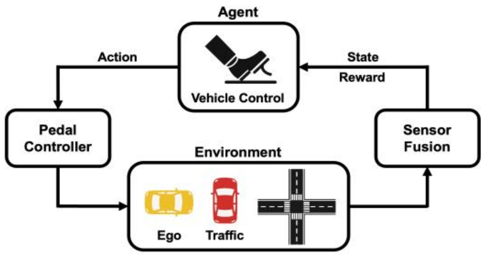


Fig. 4 The proposed reinforcement learning model

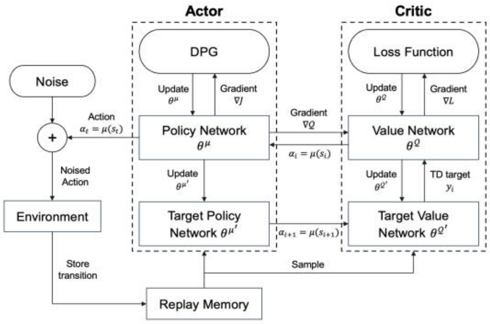


Fig. 5 DDPG architecture

Table 1 Reinforcement Learning observation

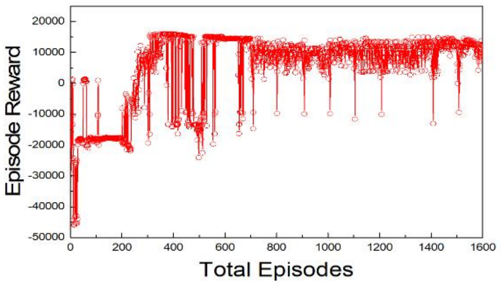
Observation	
u_{brake}	Brake Pedal
n_{scan}	LiDAR Sensor
d_x	Object Relative Distance
$v_{y1}, v_{y2}, v_{y3}, v_{y4}, v_{y5}$	Object Relative Velocity
T_s	Simulation Time
v_x	Ego Velocity
a_x	Ego Acceleration
c	Collision

Table 2 Hyperparameters of DDPG

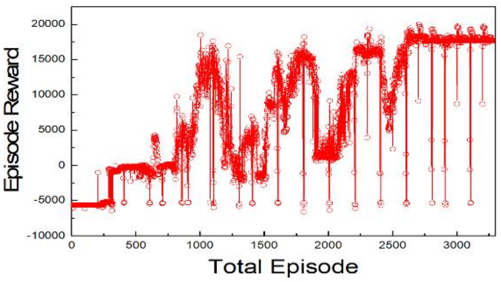
Hyperparameters		values
Actor	Learn rate	1×10^{-4}
	Gradient threshold	1
Critic	Learn rate	1×10^{-3}
	Gradient threshold	1
Agent	Sample time	0.05
	Experience buffer length	1×10^{6}
	Discount factor	0.99
	Minibatch size	64
	Noise varience	0.5
	Decay rate of noise variance	5×10^{-4}

3. Simulation Results

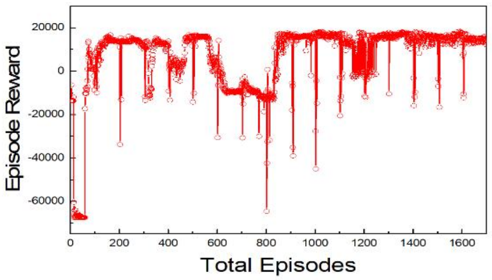
Training was performed for three intersection scenarios, and the average reward converged in all cases, as shown in Figure 6. As shown in Figure 6, the SCP/L scenario exhibited convergence at approximately 1,700 episodes, LTAP/LD converged at around 3,300 episodes, and RTAP achieved convergence within about 2,000 episodes. In the early stages of training, the reward values showed large fluctuations, but these variations gradually decreased as the number of episodes increased. In the steady-state region, the average rewards showed no significant differences across scenarios. The brake input profiles are presented in Figure 7. In the SCP/L and LTAP/LD scenarios, presented in Figure 7(a) and (b), respectively, a strong braking command occurred immediately after the opposing vehicle entered from the left, allowing the ego vehicle to stop safely without collision. In the RTAP scenario, as shown in Figure 7(c), an initial braking occurred at the early stage of risk detection, followed by a secondary braking as the opposing vehicle approached the intersection. In all three cases, the braking actions corresponded precisely to the timing of the detected hazards, and the trained policy successfully performed stable braking responses under varying environmental conditions. Figure 8 shows the variations in displacement, velocity, and acceleration under the 80 km/h initial speed condition for all three scenarios. These kinematic profiles quantitatively demonstrate the braking performance during the entire driving sequence. The results confirm that the ego vehicle was able to decelerate smoothly and come to a stable stop in each case. In addition, a speed variation test was performed by adjusting the initial speed from 60 to 90 km/h in 5 km/h increments to further evaluate the robustness of the trained policy.



(a) SCP/L



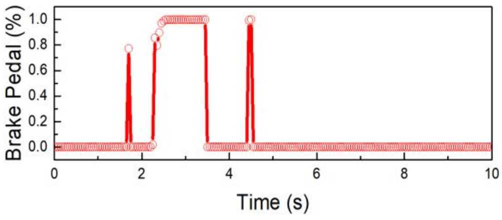
(b) LTAP/LD



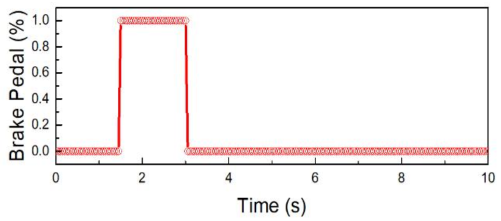
(c) RTAP

Fig. 6 Training results

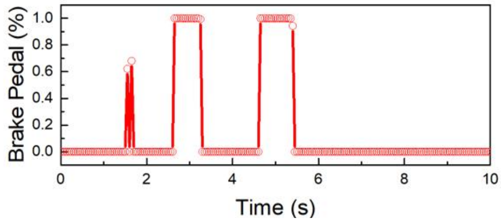
Figures 9, 10, and 11 present the velocity profiles for different initial speeds in the SCP/L, LTAP/LD, and RTAP scenarios, respectively. In all scenarios and under all test conditions, no collisions were observed. As the initial speed increased, the agent initiated braking earlier and at a greater leading distance, demonstrating adaptive behavior to maintain safety. Furthermore, Tables 3, 4, and 5 summarize the quantitative performance metrics including braking distance, minimum distance, and collision occurrence for each scenario. Across all speed conditions and scenarios, the trained policy consistently avoided collisions and maintained safe minimum distances. These results indicate that the proposed reinforcement learning based braking strategy generalizes well across varying initial speeds and diverse intersection situations.



(a) SCP/L

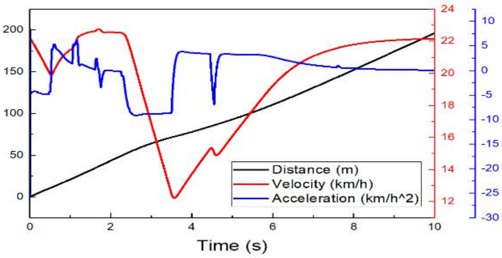


(b) LTAP/LD

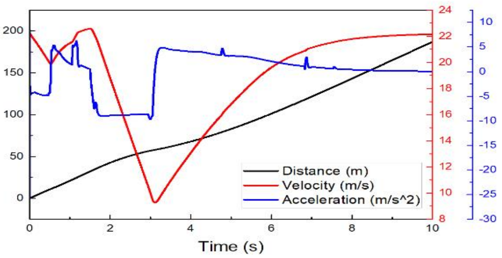


(c) RTAP

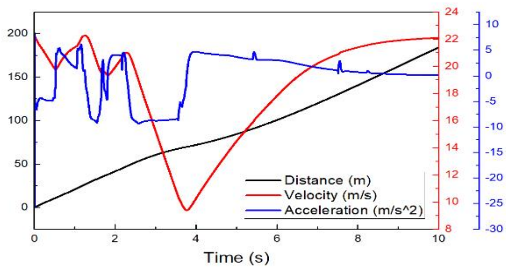
Fig. 7 Brake pedal input profiles



(a) SCP/L

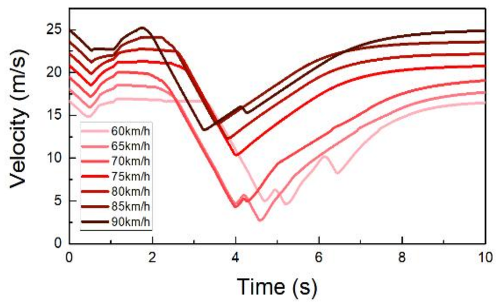


(b) LTAP/LD

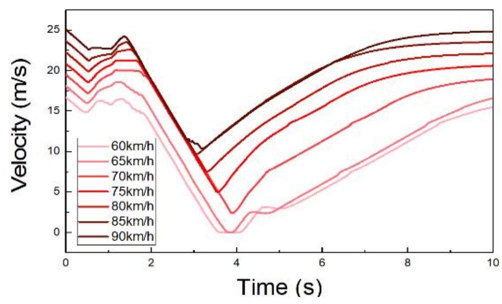


(c) RTAP

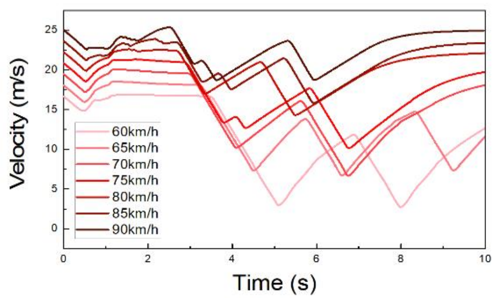
Fig. 8 Kinematic information(80km/h)



(a) SCP/L



(b) LTAP/LD



(c) RTAP

Fig. 9 Speed test result

Table 3 Speed test summary (SCP/L)

Speed	Braking Distance	Min Distance	Collision
60 km/h	18.55 m	9.77 m	No
65 km/h	26.32 m	17.89 m	No
70 km/h	26.62 m	15.91m	No
75 km/h	24.23 m	3.34 m	No
80 km/h	25.06 m	3.06 m	No
85 km/h	25.37 m	2.42 m	No
90 km/h	28.34 m	7.04 m	No

Table 4 Speed test summary (LTAP/LD)

Speed	Braking Distance	Min Distance	Collision
60 km/h	15.31 m	4.54 m	No
65 km/h	15.24 m	3.45 m	No
70 km/h	15.20 m	4.03 m	No
75 km/h	15.48 m	4.74 m	No
80 km/h	14.89 m	5.37 m	No
85 km/h	24.84 m	4.79 m	No
90 km/h	23.70 m	5.86 m	No

Table 5 Speed test summary (RTAP)

Speed	Braking Distance	Min Distance	Collision
60 km/h	15.31 m	4.54 m	No
65 km/h	15.24 m	3.45 m	No
70 km/h	15.20 m	4.03 m	No
75 km/h	15.48 m	4.74 m	No
80 km/h	14.89 m	5.37 m	No
85 km/h	24.84 m	4.79 m	No
90 km/h	23.70 m	5.86 m	No

4. Conclusion

In this study, a reinforcement learning based autonomous emergency braking system was applied to unexpected intersection scenarios, and its performance was verified through simulation. The DDPG algorithm was employed to learn a continuous braking control policy, and three intersection scenarios were implemented in a virtual environment integrated with IPG CarMaker and MATLAB/Simulink. The training results showed that the reward values converged stably across all scenarios, and the trained agent successfully avoided collisions through appropriate braking actions when hazardous situations occurred. Furthermore, in the speed variation tests, no collisions were observed under any conditions, confirming that the proposed method maintained consistent and reliable braking performance across different driving speeds. Although DDPG was effective for continuous braking control in this study, relying solely on a single reinforcement learning algorithm limits the ability to evaluate how different methods might influence safety, braking smoothness, and convergence properties. Future work will therefore apply alternative continuous-control reinforcement learning algorithms such as Proximal Policy Optimization (PPO) and Soft Actor–Critic (SAC). PPO will be explored to assess whether its stable on-policy updates can improve braking consistency under frequent distribution shifts in intersection scenarios, while SAC will be examined to determine whether its entropy-regularized objective enhances robustness and responsiveness under uncertain and highly dynamic traffic conditions. Comparative analysis of braking distance, minimum distance to obstacles, collision rate, and comfort-related metrics will help identify reinforcement learning strategies that best balance safety and ride comfort. Furthermore, the current simulation environment considers only vehicle-to-vehicle interactions and does not include more unpredictable and vulnerable road users such as pedestrians, bicycles, motorcycles, or scooters. These agents often exhibit irregular and noncompliant behaviors, posing additional challenges for autonomous emergency braking systems. Future research will expand the simulation environment to incorporate heterogeneous road users with diverse motion patterns and right-of-way behaviors, as well as environmental factors such as adverse weather, nighttime illumination, and sensor uncertainty. Incorporating these elements into the state representation and training scenarios will enable a more realistic evaluation of braking policies and support the development of AEB strategies that generalize effectively to complex real world conditions.

Acknowledgement

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